Mert Albaba

Webpage | E-Mail | Google Scholar | LinkedIn

How can we enable robots to autonomously learn new skills? I build robotics approaches that combine reinforcement learning, world models, and vision-language-action models to teach humanoid robots complex behaviors.

Research Interests: Robotics · Vision-Language-Action Models (VLAs) · World-Action-Models (WAMs) · Reinforcement Learning

7+ years of research in reinforcement learning · 10 first-author papers

EDUCATION

ETH ZÜRICH & MAX PLANCK INSTITUTE | PhD in Computer Science — July 2022 – Present

ELLIS PhD Student · CLS Fellow · Advisors: Michael Black and Andreas Krause

Research on humanoid robots, robot learning, and reinforcement learning.

BILKENT UNIVERSITY | MSc and BSc in Electrical Engineering — 2015 – 2021

Research Fellow · Research Award

RESEARCH EXPERIENCE

META | Research Intern | London — September 2026 – February 2027

- Research on world-action-models and self-supervised learning.

VESOMA | Research Intern | Munich — May 2026 – September 2026

- Research on generalist robot policies, VLAs and WAMs, supervised by Martin Riedmiller.
- Created **VioLA** , a generalist humanoid policy learned mostly from human data, enabling zero-shot locomotion and manipulation.

MESHCAPADE | Machine Learning Research Intern | Zürich — July 2025 – December 2025

- Research on humanoid robots and human motion generation.
- Worked on a generalist model for text-to-humanoid motion generation.

ETH ZÜRICH & MAX PLANCK INSTITUTE | Scientific Researcher | Zürich — July 2022 – Present

- Proposed **NIL** (**CVPR 2026**), the **first data-free imitation learning** method: leverages pretrained video diffusion models to synthesize robot demonstrations, then distills physically plausible locomotion policies purely from the synthetic data.
- Developed **RILe**, a trainer–student inverse reinforcement learning framework **outperforming prior state-of-the-art by 20%** on humanoid locomotion benchmarks.
- Designed **GC-TTT**, a self-supervised test-time adaptation method for offline goal-conditioned RL that selects relevant experience at inference to **improve policy generalization without full retraining**.

SYSTEMS LAB, BILKENT UNIVERSITY | Researcher — March 2017 – July 2022

- Research on integration of **reinforcement learning** with **game theory** for modeling complex multi-agent human behaviors.
- Published 4 first-author papers including a journal review article.

SKILLS

EXPERTISE: • Robotics • Reinforcement Learning • VLAs & WAMs • Humanoid Robots

TECHNICAL SKILLS: • Python • C++ • C | • JAX • PyTorch | • MuJoCo (MJX) • Isaac Lab • SLURM / HPC

SELECTED PUBLICATIONS

- Link **VioLA: Learning Generalist Humanoid Control Policies from Human Data** *arXiv 2026*
  Mert Albaba*, Jens Beißwenger*, Anna Manasyan* et al. 'VioLA: Learning Generalist Humanoid Control Policies from Human Data.' arXiv preprint, 2026.
- Link **NIL: No-data Imitation Learning by Leveraging Pre-trained Video Diffusion Models** *CVPR 2026*
  Albaba, Mert, et al. 'NIL: No-data Imitation Learning.' Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2026. Spotlight talk at 2nd Workshop on Agents in Interaction, from Humans to Robots.
- Link **GC-TTT: Test-time Offline Reinforcement Learning on Goal-related Experience**. *ICML 2026*
  Marco Bagatella*, Mert Albaba*, et al. 'Test-time Offline Reinforcement Learning on Goal-related Experience.' Proceedings of the 43rd International Conference on Machine Learning, 2026.
- Link **Moving by Looking: Towards Vision-Driven Avatar Motion Generation**. *Under Review*
  MarkosDiomataris, Berat Mert Albaba, Giorgio Becherini, Partha Ghosh, Omid Taheri, and Michael J. Black. 'Moving by looking: Towards vision-driven avatar motion generation.' arXiv preprint, 2025.
- Link **RILe: Reinforced Imitation Learning**. *Robot Learning Workshop at ICLR 2025*
  Albaba, Mert, et al. "Rile: Reinforced imitation learning." arXiv preprint arXiv:2406.08472 (2024).

ACCOMPLISHMENTS

- Ranked **top 0.0001%** (15th among 2 million students) in the National University Graduate Examination